# Saurabh Surashe

Member of Technical Staff III (R&D) - Mavenir Systems

*+91-7276561155* || saurabhsurashe@gmail.com || *LinkedIn*


## Professional Summary


Software Engineer with 4 years of experience in C/C++ and Python, specializing in low-latency, real-time networking protocols, wireless L2/L3 stack development (RAN/O-RAN), embedded systems. Proven expertise in designing scheduling algorithms, shared memory IPC modules. Experienced in integrating edge AI inference models into real-time RAN architectures to optimize link adaptation and random access channels. Adept in handling Field/Customer Issues.


## Core Competencies

**Systems Programming**: C, C++11/14/17, Python, STL, Multithreading, Concurrency, Socket Programming, Shared Memory IPC, Linux, Data Structures, Algorithms
**AI and Machine Learning**: TensorFlow, Deep Learning, NLP, Transformers, Embeddings, Semantic Search, Vector Similarity Search, Feature Engineering
**Performance and Debugging**: GDB, VTune, Wireshark, QXDM, Amarisoft, Profiling, Root Cause Analysis
**Development Practices**: Agile Development, System Design, Software Architecture, Git
**Telecom and Networking**: NB-IoT, NTN, TCP/IP, UDP, MAC, RRC, 3GPP Release 15/17

## Professional Experience

**Mavenir Systems - Member of Technical Staff III (R&D)** March 2026 - Present
*NTN and TN NB-IoT (Non-Terrestrial and Terrestrial Networks)*

- Co-designed and implemented a **high-performance Data Collection Module** for real-time Edge AI model integration with base stations, optimizing RACH Inference handling and Uplink Link Adaptation.
- Participated in customer issue investigation, debugging, validation, and deployment activities.

**Mavenir Systems - Member of Technical Staff II (R&D)** February 2025 - February 2026
*NTN and TN NB-IoT*

- **Re-architected** the MAC scheduler to support efficient multi-UE scheduling, boosting base station throughput by **30%** and resource block utilization by **60%**.
- Implemented over **60+ Performance Counters** spanning RRC, MAC, and RLC layers, enabling detailed telemetry collection, customer debugging, and root-cause analysis.

**Mavenir Systems - Member of Technical Staff I (R&D)** July 2022 - January 2025
*NTN and TN NB-IoT, LTE-M Development*

- Designed and developed a low-latency, **Python-based Shared Memory (SHM) IPC module** for satellite ephemeris synchronization, reducing data transfer overhead and improving multi-process synchronization.
- Enhanced Mavenir's proprietary Uplink Scheduling Algorithm, reducing transmission delays in satellite communication (NTN) systems and boosting uplink throughput **by 40%**.
- Contributed to **full-cycle** LTE-M eNodeB development **(design to deployment)** on 4G eNB infrastructure.
- **Co-designed** Uplink MAC and MPDCCH schedulers, optimizing uplink performance and hardware resource allocation.

## Projects

***AI-Powered Personal AI Job Scraper & Match Engine:***
- Engineered a local Python FastAPI & SQLite pipeline to ethically scrape and store job openings from 10+ major tech portals using custom Workday, Eightfold, and JSON API crawlers.
- Built a local, offline NLP service (all-MiniLM-L6-v2 via PyPI SentenceTransformers) using an overlapping chunking algorithm to embed and match whole-resume vectors against job listings.
- Implemented a 50/50 weighted title-to-description similarity engine leveraging high-performance NumPy vector cosine similarity calculations.
- Integrated asynchronous background task loops to dispatch sorted email alerts (smtplib/MIME) featuring boundary-safe skill tags, location-priority sorting, and experience warning badges.

***Malicious web page detection using Machine and Deep Learning:***
- Developed a cybersecurity solution to classify malicious and benign URLs using machine learning and deep learning techniques.
- Engineered lexical and NLP-based features from URL datasets containing 10,000+ samples.
- Built and evaluated classification models using Scikit-learn, TensorFlow, and Keras.

## Publications & Awards

Mavenir **Spot-Light Awards** for excellence in Research and development.
Co-authored: **"Privacy and Security Comparison of Web Browsers: A Review"**, AINA 2022 (*SpringerLink*)
Co-authored: **"Enhancing Cyber Security: Malicious web page detection using Machine and Deep Learning"**, AINA 2025 (*SpringerLink*)

## Education

**National Institute of Technology Surathkal** Karnataka
*Master of Technology (M.Tech) in Computational Data Science* *June 2020 - May 2022*

**MIT College of Engineering Pune** Maharashtra
*Bachelor of Engineering (B.E.) in Mechanical Engineering* *June 2015 - May 2019*